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Chinese Pharmaceuticals (HK) Limited: Effective Forecasting for Optimal Inventory Management

Accurate forecasts are vital to our business because they drive all of our production and distribution decisions.[1]

Jason Kwok, general manager, Chinese Pharmaceuticals

Chinese Pharmaceuticals (HK) Limited was a young and successful Hong Kong company supplying Chinese herbal medicine products to the Hong Kong market. The company's largest customer was Mannings, the leading drugstore chain in Hong Kong. While the company had expanded its product portfolio, which included several brands, its best-selling product remained Noto37, a Chinese herbal medicine used to control cholesterol and blood pressure levels.

Jason Kwok, general manager of Chinese Pharmaceuticals, was very pleased with the strong sales of its marquee Noto37 product. Those strong sales, though, were creating other challenges for the company. In the previous week, Jason had learned that Noto37 was out of stock again. Retail space in Hong Kong was precious, and retailers like Mannings could not afford to have empty shelves. If his company could not get product delivered to Mannings shortly, Mannings' purchasing manager might be forced to stock other herbal medicine products. Jason needed to review the supply chain for Noto37 and determine some ways to first improve the company's sales forecasts and then implement better inventory management practices.

Chinese Pharmaceuticals (HK) Limited

Chinese Pharmaceuticals followed the trajectory of many other Hong Kong companies. It began as a small, family-owned trading company in 1998, dealing primarily in Chinese herbs. As margins of trading companies continued to be squeezed and the Hong Kong government encouraged the development of higher-

[1] Interview by author, 18 May 2012.

Joseph S. Fernandez prepared this case under the supervision of Professor Ronald Lau solely as a basis for class discussion. The authors have disguised certain data to protect confidentiality. Cases are written in the past tense; this is not meant to imply that all practices, organizations, people, places or facts mentioned in the case no longer occur, exist or apply. Cases are not intended to serve as endorsements, sources of primary data, or illustration of effective or ineffective handling of a business situation.

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value industry sectors, Chinese Pharmaceuticals decided to develop its own, proprietary Chinese herbal medicine products. Through the government endowed Hong Kong Science and Technology Park, it established its first research and development office in 2007. Thus began its transformation from strictly a trading entity to a developer and marketer of its own branded range of Chinese herbal medicine products. Its first signature product was Noto37, an herbal remedy whose primary ingredient was Notoginseng, known to be effective in controlling cholesterol and also in improving cardiovascular health.

Products

Over the years, the company had gradually developed additional products, such as Gaba, Banana Detox, and Water-Bean. Each of these new products addressed different ailments. Gaba, for example, was developed to address insomnia; Banana Detox was developed as an herbal colon cleansing and detoxification regimen; and Water-Bean was targeted at the booming slimming and weight loss market. Nevertheless, Noto37 remained the star performer for the company and accounted for the majority of the company's sales and revenues as much as 80% during the peak winter sales period.[2]

Markets

As a Hong Kong company with well-established roots in the local market, Chinese Pharmaceuticals decided to make Hong Kong the main focus of its sales and marketing efforts. The Hong Kong market accounted for as much as 95% of the company's sales. The other market for the company was Macau, with sales having also begun to the Chinese community in the United States through an agent.[3]

The Hong Kong Market for Chinese Herbal Medicines

Main Brands in the Market

Traditional Chinese medicines had been used by people of Chinese origin for hundreds of years, with the earliest written documentation dating to almost 2,000 years ago.[4] These traditional medicines, therefore, had always been a part of Hong Kong society. As Hong Kong society developed, traditional outlets for Chinese medicines, such as well-established players like Eu Yan Sang, adapted their business models. In addition to selling Chinese herbal medicines in traditional loose form, they also sold products in capsule form and in eye-catching retail packaging. While the majority of their retail products had been sold under their house brand—Eu Yan Sang—they also developed several sub-brands.

While Eu Yan Sang had been a long-term player in the market, newer entrants included PuraPharm, which developed an extensive range of consumer Chinese herbal medicine products for almost every possible ailment. PuraPharm's products had been sold under the PuraPharm and PuraPharm Gold brands, as well as a multitude of other brands. Several other players had also entered the increasingly crowded market for Chinese herbal medicines in Hong Kong.

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[2] Interview by author, 18 May 2012.
[3] Ibid.
[4] Hong, Francis F. (2004), "History of Medicine in China When Medicine Took an Alternative Path," *McGill Journal of Medicine*, pp. 79-84.

Chinese Pharmaceuticals' Market Share

Chinese Pharmaceuticals confronted what was not necessarily a saturated market, but an increasingly competitive marketplace that featured companies with well-established brands and newer entrants eager to capitalize on the trend of packaged, easy-to-consume Chinese herbal medicines. Nevertheless, with its initial product, Noto37, Chinese Pharmaceuticals had been able to capture 40% of the market share for Chinese herbal medicines targeting cholesterol, blood pressure, and related cardiovascular ailments. It had done so through effective marketing campaigns that included special promotions in a different mall each week, and also the use of local celebrity spokespersons.[5]

Consumer Awareness of Health Issues

The development of the Chinese herbal medicine market in Hong Kong was also affected by greater awareness of major health issues through information obtained from readily accessible media outlets, including print, television, and the Internet. The Hong Kong government, for example, provided statistics from a 2009 study of the leading causes of death in Hong Kong on its Brand Hong Kong Web site.[6] Two of the top four leading causes of death were diseases of the heart and stroke, both of which could be impacted by high cholesterol levels. Therefore, the launch of Chinese Pharmaceuticals' Noto37, which addressed problems of the heart and cholesterol, coincided well with the increased awareness of such ailments among consumers in Hong Kong.

Supply Chain for Noto37

China Vendor

The Notoginseng ingredient used in Chinese Pharmaceuticals' flagship product, Noto37, originated from Yunnan Province in China. Chinese Pharmaceuticals had an established supplier of the Notoginseng in Yunnan that was able to provide the raw material in a “super fine” form, enabling it to be prepared into ready-to-consume tablets or capsules. While the Notoginseng was available year-round, it was particularly abundant in the spring and winter. The excess supply during these times of the year also resulted in lower prices. Due to cash-flow constraints, Chinese Pharmaceuticals was not always able to take advantage of the lower prices and preferred instead to purchase regular quantities year-round. From the time of order placement, the supplier was usually able to deliver the Notoginseng in super fine form within four to six weeks.

Testing of Raw Materials

Prior to taking delivery of new stock of Notoginseng, Chinese Pharmaceuticals conducted testing of random samples at independent labs in China to ensure that excess metals or toxins were not present. It also conducted a water content test, as excess water could contribute to bacteria growth. Such testing normally took no more than seven to nine days.

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[5] Interview by author, 18 May 2012.

[6] Brand Hong Kong, Health Information Library, Government of Hong Kong, “Leading Causes of All Deaths,” http://www.healthyhk.gov.hk/phisweb/plain/en/healthy_facts/disease_burden/major_causes_death/major_causes_death, accessed 17 June 2012.

Hong Kong Manufacturer

Upon successful testing of the samples, new deliveries of Notoginseng were made to Chinese Pharmaceuticals' leased warehouse facility in Hong Kong. Stock was held in the warehouse until it was ready to be delivered to the Hong Kong based contract manufacturer, which then required up to two months to complete production of the Noto37 product. Newly manufactured Noto37 was delivered to the same warehouse facility.

The total lead time, therefore, from the time Chinese Pharmaceuticals placed its initial order with its Yunnan supplier to having stock of new product to deliver to its local customers, was at least 90–100 days [see **Exhibit 1**].[7]

Weekly Deliveries of Noto37

Chinese Pharmaceuticals worked with a local transportation company to arrange delivery of new stock of Noto37 to its local customers, usually once or twice a week. In addition to deliveries to its largest customer, Mannings, which operated 300 stores in Hong Kong,[8], deliveries also had to be coordinated to Mannings' main rival, Watson's, with over 180 outlets in Hong Kong and Macau,[9], as well as to over 500 local, neighborhood pharmacies throughout Hong Kong. Besides delivering new stock to the retail outlets, Chinese Pharmaceuticals also had to intermittently collect old or damaged stock from the stores. These would usually be minimal, and the company ordinarily simply disposed of return stock.

Inventory Management

Seasonality of Sales

The winter months leading up to the Chinese New Year Festival were the clear peak season for sales of Noto37, with sales increasing as much as 50% above average during the cold-weather season. Sales would then, ordinarily, decrease by as much as 50% below average during the subsequent spring season, while stabilizing toward their annual average during the summer and autumn months.[10]

Management of Inventory at the Hong Kong Warehouse

As the company did not have its own warehouse facilities, but leased out storage space, it had to carefully manage its storage needs. Leading up to its peak winter sales period, the company would secure additional warehouse space to accommodate its higher inventory levels. Ideally, the company would seek to maintain at most three months' supply of stock, as this was also the capacity of its leased warehouse space for finished products. Reordering was triggered when stock levels shrunk to as little as 1.5 months' supply. The reality, however, was that various factors impacted the stock levels, including the biweekly deliveries of different quantities to almost 1,000 retail outlets, intermittent delays in supply of new stock from the manufacturer,

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[7] Interview by author, 18 May 2012.
[8] Mannings, "Store Network," http://www.mannings.com.hk/eng/store_location.html, accessed 17 June 2012.
[9] Watson's Your Personal Store, "About Us," http://www.watsons.com.hk/webhk/HistoryWatsons.html, accessed 17 June 2012.
[10] Interview by author, 18 May 2012.

and drought conditions in Yunnan that could impact supply of the key Notoginseng. Company sponsored weekly promotions plus the company's own intermittent promotions further contributed to fluctuations in demand and, consequently, on stock levels.

While stock levels could be expected to fluctuate during the course of the year, Jason realized that an out-of-stock situation was certainly not acceptable, especially for the company's best-selling product and primary revenue driver. He struggled, therefore, to establish a more predictable means of managing the inventory.

Inventory management for the company had been predicated more on circumstance than any systematic inventory management practice. For example, the company's inventory position[11] was likely affected more by such factors as the available warehouse space leased and less by any formal assessment and analysis of inventory positions in prior months. Jason realized that it would be prudent to incorporate more effective forecasting and more formalized inventory management practices. Jason vaguely recalled inventory management and forecasting exercises he had undertaken during one of his university courses many years earlier, and wondered if such techniques would be appropriate for his company. As the company had recently hired an intern who was completing his MBA, Jason decided to assign the intern with the task of determining more appropriate inventory management and forecasting tools the company could utilize to improve its inventory management practices and avoid his current predicament of no stock.

Forecasting Tools

The Intern's Recommendation

Provided with access to the company's sales history for the previous three years [see **Exhibit 2**], the intern considered the options. He realized that Jason required an urgent assessment of the company's sales data and a recommendation for a forecasting tool that could be implemented in the near term. With a better forecasting model, Jason would then be in a better position to consider revised inventory management practices based on an analysis of past demand plus current demand, as well as forecast demand for the Noto37. In reviewing the sales data, the intern realized that different variables could have impacted the data over the previous three years. So, how could he best use this historical data to foresee future sales? One method the intern identified was simple exponential smoothing.[12]

Importing the data into an Excel worksheet and utilizing Excel tools, he proceeded to carry out the exponential smoothing exercise in the hopes of delivering useful forecasting results [see **Exhibit 3** for simple exponential smoothing results].

Pleased with his efforts, the intern presented the results to Jason in the hopes that the information would be useful to Jason in improving the company's forecasting and inventory management challenges.

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[11] Inventory position = Inventory level + On order inventory; Inventory level = On-hand inventory -– backorders.

[12] Exponential smoothing was the most used of all forecasting techniques in inventory ordering decisions for retail and wholesale companies. The next forecast is updated with the current forecast and actual sales data. A crucial element in the exponential smoothing forecasting model is the smoothing constant alpha. This smoothing constant determines the level of smoothing and degree of reaction needed when there is a difference between forecast and actual sales data. The value of alpha is set between 0 and 1.

F_{t+1} = alpha * S_t + (1 – alpha) * F_t, or alternatively

F_{t+1} = F_t + alpha * (S_t – F_t) where F_t = forecast sales for period t and S_t = actual sales for period t.

Concerns about Accurate Forecasting

Immediately after his meeting with Mannings, Jason had obtained an update on deliveries of Notoginseng from his supplier in Yunnan and convinced the supplier to use extra workers to gather and process an additional supply of Notoginseng for his contract manufacturer in Hong Kong. He had also confirmed that testing for metals and toxins could be completed in half the normal time. He had also then pressed the manufacturer in Hong Kong to jump the production queue so that he could get an initial small delivery of product in two weeks' time, with a larger quantity to then be delivered in thirty days. He hoped this would appease the purchasing managers at Mannings and other retail outlets.

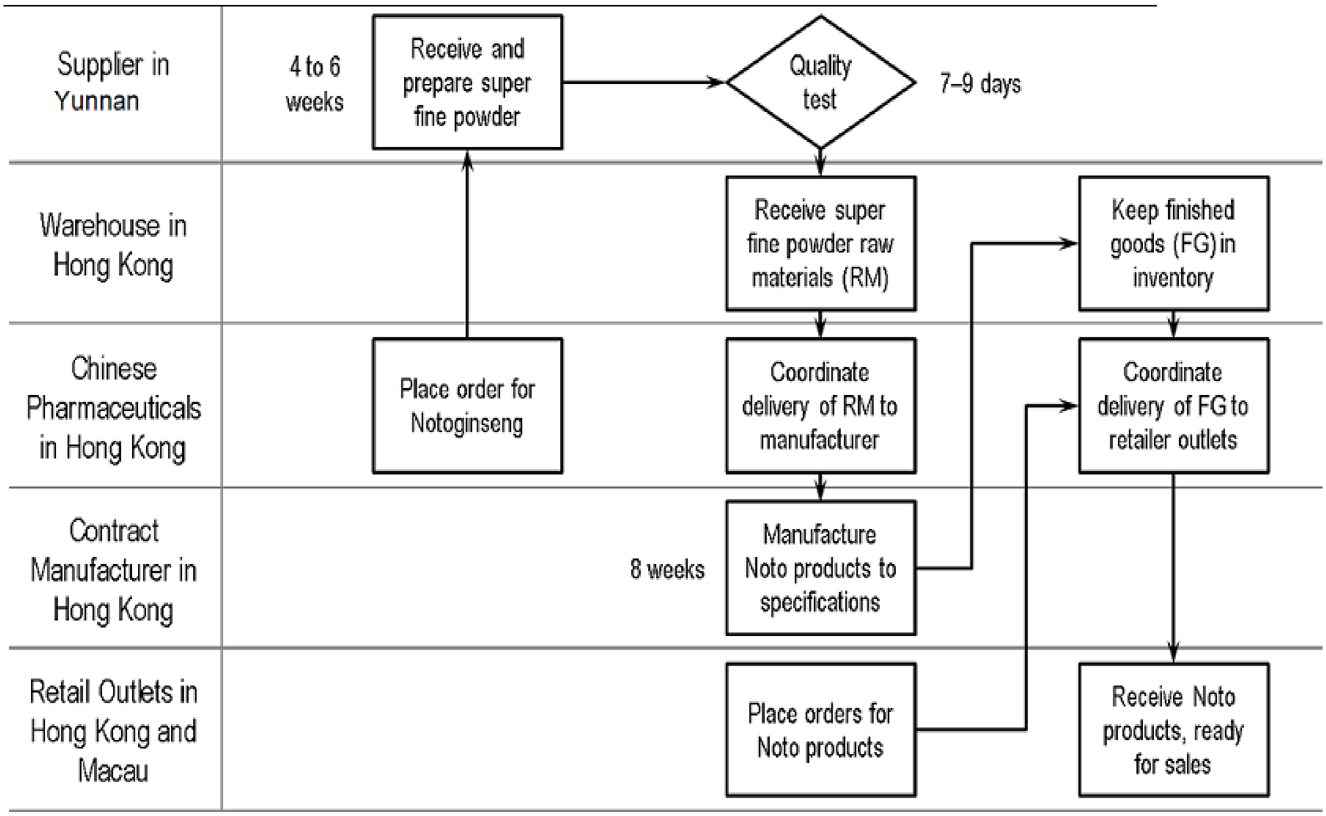
In addition, however, Jason reviewed the work done by his new intern. He was pleased with the quick turnaround on the part of the intern. Nevertheless, as Jason reviewed his company's sales data for the previous three years and the forecasts the intern had compiled, he wondered how accurate the calculations were. Might there be some additional forecasting methods that could produce more accurate results? Jason was keen to establish the right forecasting model for the company, which he hoped would then lead to improved inventory management of the company's best-selling Noto37, and fewer challenging meetings with purchasing managers.

Conclusion

Chinese Pharmaceuticals was a Hong Kong success story in more ways than one. Transforming itself from a trading company to a researcher, developer, and marketer of award -winning and market-leading Chinese herbal medicine products was no small task. Much effort and investment had gone into building up the company. The company's best-selling product, Noto37, was a good reflection on the company that developed it. It was a consumer-friendly Chinese herbal medicine product in capsule form that was promoted by celebrities and gladly given a prominent space on the shelves of the leading drugstore chain in Hong Kong, Mannings.

With growth, however, had come some growing pains. Most critical for the company and its general manager, Jason Kwok, was a misalignment between production and delivery of the Noto37 product and the high demand for the product in the market. This resulted in an out-of-stock situation for Noto37 and an unhappy purchasing manager at the company's biggest account, Mannings. One means to rectify the situation was to enhance the company's ability to improve the sales forecasts for its best-selling item. With an improved ability to forecast demand, the company could undertake the necessary adjustments in its supply chain to ensure that adequate inventory of product was always maintained at the local warehouses it leased. It was critical, therefore, that the forecasting method used would provide Jason with the most accurate estimates of future demands.

EXHIBIT 1: SUPPLY CHAIN FOR NOTO37 PRODUCTS



Source: provided by company

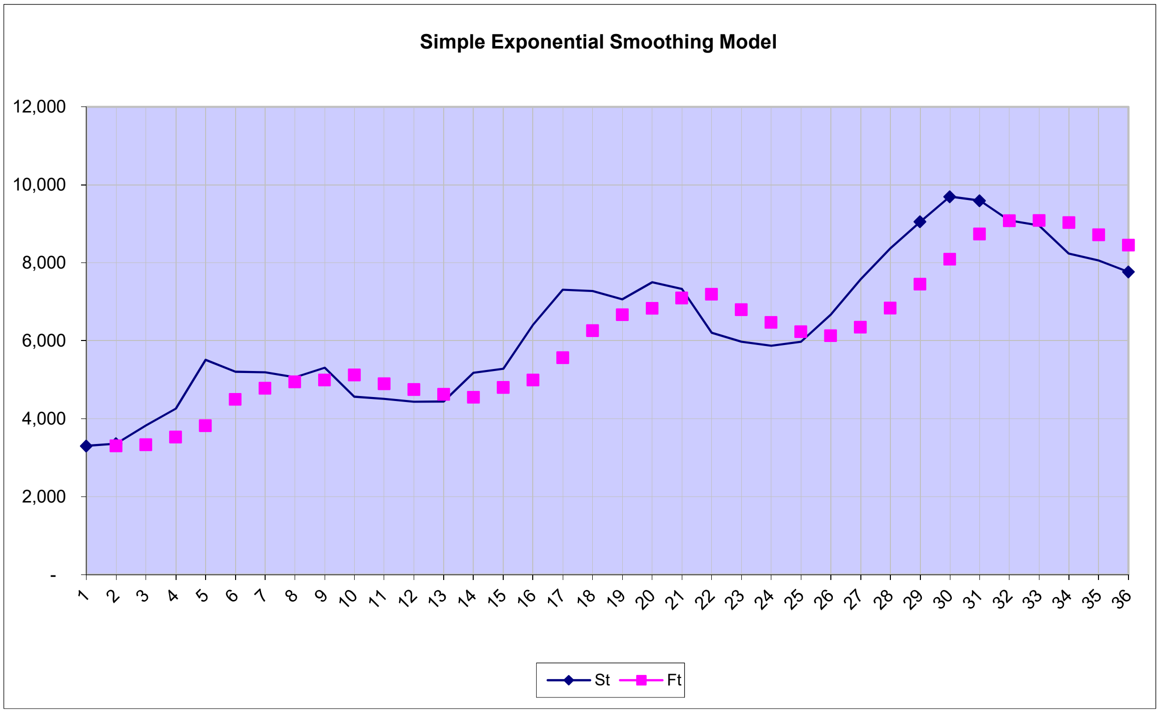
EXHIBIT 2: SALES AND FORECAST DATA

MM-YY	Period t	Sales S_t	Forecast F_t
Jul-09	1	3,303	
Aug-09	2	3,360	3,303
Sep-09	3	3,828	3,326
Oct-09	4	4,257	3,527
Nov-09	5	5,508	3,819
Dec-09	6	5,205	4,494
Jan-10	7	5,190	4,779
Feb-10	8	5,058	4,943
Mar-10	9	5,307	4,989
Apr-10	10	4,563	5,116
May-10	11	4,512	4,895
Jun-10	12	4,434	4,742
Jul-10	13	4,440	4,619
Aug-10	14	5,178	4,547
Sep-10	15	5,277	4,800
Oct-10	16	6,411	4,991
Nov-10	17	7,308	5,559
Dec-10	18	7,275	6,258
Jan-11	19	7,065	6,665
Feb-11	20	7,497	6,825
Mar-11	21	7,326	7,094
Apr-11	22	6,207	7,187
May-11	23	5,976	6,795
Jun-11	24	5,874	6,467
Jul-11	25	5,970	6,230
Aug-11	26	6,666	6,126
Sep-11	27	7,575	6,342
Oct-11	28	8,367	6,835
Nov-11	29	9,051	7,448
Dec-11	30	9,696	8,089
Jan-12	31	9,594	8,732
Feb-12	32	9,084	9,077
Mar-12	33	8,955	9,080
Apr-12	34	8,235	9,030
May-12	35	8,055	8,712
Jun-12	36	7,767	8,449

Note: The alpha value = 0.4 and initial forecast $F_2 = S_1 = 3{,}303$ units.

Source: Disguised data provided by Chinese Pharmaceuticals (HK) Limited.

EXHIBIT 3: SALES AND FORECAST DATA IN GRAPHICAL DISPLAY



Note: S_t and F_t are the actual sales and forecast, respectively, for the time period t.

Source: Disguised data provided by Chinese Pharmaceuticals (HK) Limited.